A3-PS003-TC371

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

Dr. J. Cynthia
Cynthia@karunya.edu
Aryashree Rajkumar
aryashreerajkumar@karunya.edu.in
Yuthika G
yuthikag@karunya.edu.in

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Abstract: Problem Description

- Streaming raw junction data to a central cloud uses cellular bandwidth and pools private travel logs.
- Federated Learning keeps data on-site and shares only weight updates, but real junctions are non-IID and every round costs bytes.
- We build a federated simulator on a public PEMS04 (PeMS) traffic-flow dataset split across 5 virtual junctions. One compact GRU is trained with FedAvg and compared with centralized and local-only models.
- We test IID and non-IID splits with chronological train/test splits over 3 seeds, and study how non-IID data changes accuracy, convergence and megabytes sent.

How accurate?

MAE and RMSE for FedAvg vs centralized and local-only models, IID vs non-IID.

How costly?

Bytes up and down per round, and total MB to reach a target RMSE.

How fast?

Rounds to converge under each split. Mean ± std over 3 random seeds.

Background

Traffic-flow prediction

Forecasting vehicle counts at a junction 15–60 minutes ahead so authorities can act before congestion builds.

Federated Learning

Each junction trains on its own data; only model weights, never raw records, go to the server for averaging.

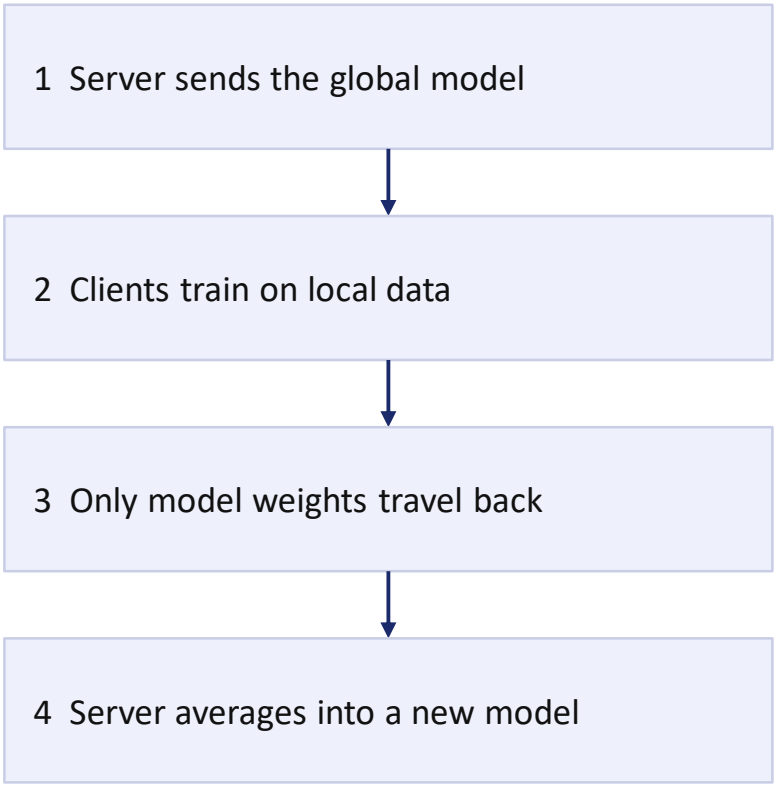
IID vs non-IID

IID: every client sees the same kind of traffic. Non-IID: a highway ramp and a school street differ, so one averaged model fits neither well.

Communication cost

Bytes up + down per client per round, times rounds. Per round ≈ 2 × clients × model size.

One federated round (FedAvg [1])



Literature & RFC Review

Work	What it contributes	How we use it / gap we address
FedAvg [1]	Weighted averaging of locally trained models	Core algorithm. We add exact byte accounting per round
DCRNN [2]	Graph-based traffic forecasting (METR-LA)	Needs centralized data. Context only; we use a compact GRU
FL traffic prediction [3]	Federated traffic-flow forecasting with a privacy focus	We foreground bytes and convergence under controlled IID / non-IID splits
FedProx [7]	Proximal term to handle heterogeneous clients	Future work; not implemented
Deep Gradient Compression [4]	Sparsified gradients cut bandwidth	Future work; not implemented
Clustered FL (IFCA) [6]	Group clients with similar data	Context only; not implemented
RFC 8446 [9]	TLS 1.3 for secure channels	Deployment note only; not simulated

Out of scope (per problem statement): formal privacy guarantees, secure aggregation [5], live cameras, city-wide deployment.

Proposed Solution

Federated simulator with controlled IID / non-IID splits and a byte meter

1

Load & split

PEMS04 flow data (307 detectors). Chronological train / val / test; scaler fit on train only.

2

Partition

Assign detectors to 5 virtual junctions: IID (random) or non-IID (grouped by flow level).

3

Local training

Each client trains a compact GRU on sliding windows for a few local epochs.

4

FedAvg + byte meter

Server averages weights. Bytes up and down are logged every round.

5

Evaluate

MAE, RMSE, rounds to target, MB to target. 3 seeds, mean ± std.

What we measure: IID vs non-IID accuracy • bytes sent per round • MB to reach a target RMSE • rounds to converge

Experiment Design and Optimization

Aspect	What we run	Why it matters
Data & split	One PEMS04 flow dataset; chronological train / val / test; scaler fit on train only	No data leakage
Clients	5 virtual junctions	Within the 3–5 client rule
IID vs non-IID	IID: random detector assignment. Non-IID: detectors grouped by flow level	Shows how heterogeneity hurts FedAvg
Non-IID gap	Difference in mean flow between clients (simple)	Relates the gap to rounds and MB to target
Methods	Persistence, local-only, centralized, FedAvg (E = 1; E = 5 optional)	Lower / upper bounds; accuracy vs bytes
Metrics	MAE, RMSE, bytes per round, total MB, rounds to target; seeds 0, 1, 2	Fair, reproducible comparison

Optional if time permits: E = 5 run, worst-client RMSE. Future work: quantization, FedProx [7]. Results reported as measured; no gains assumed.

Solution Architecture & Design

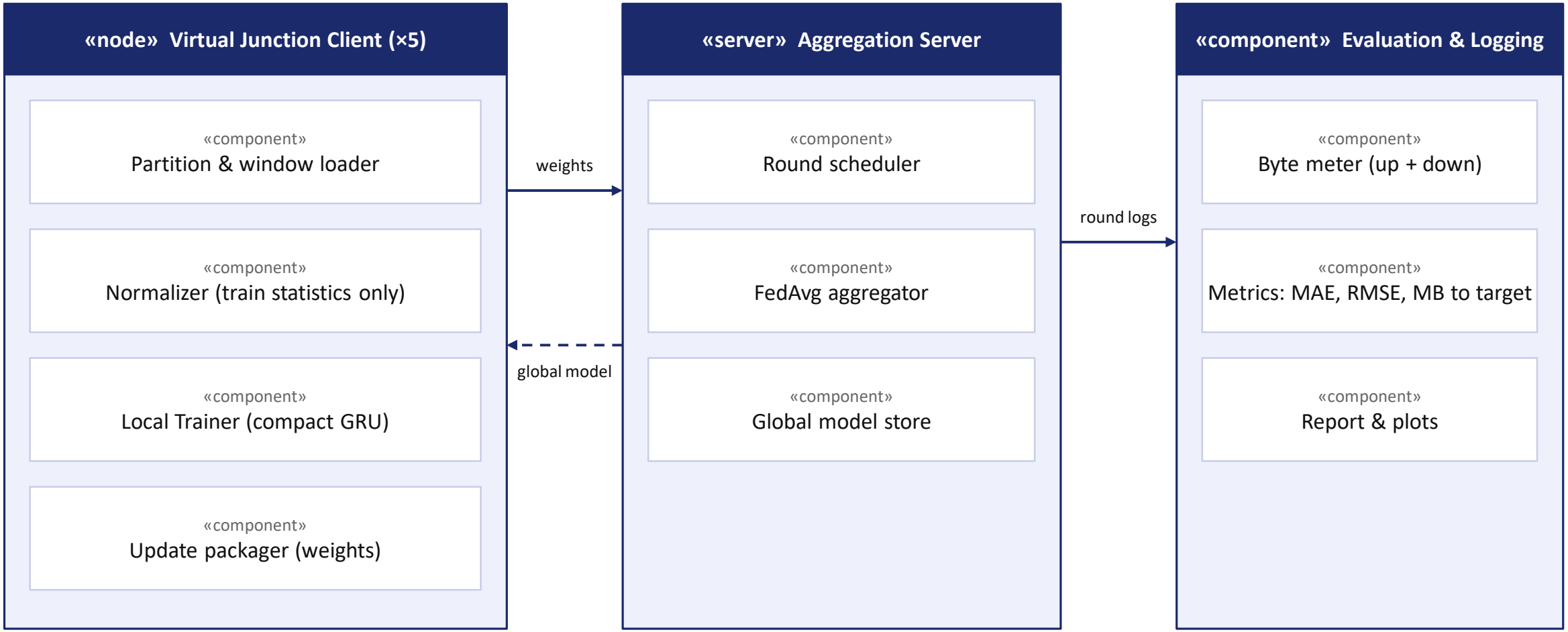
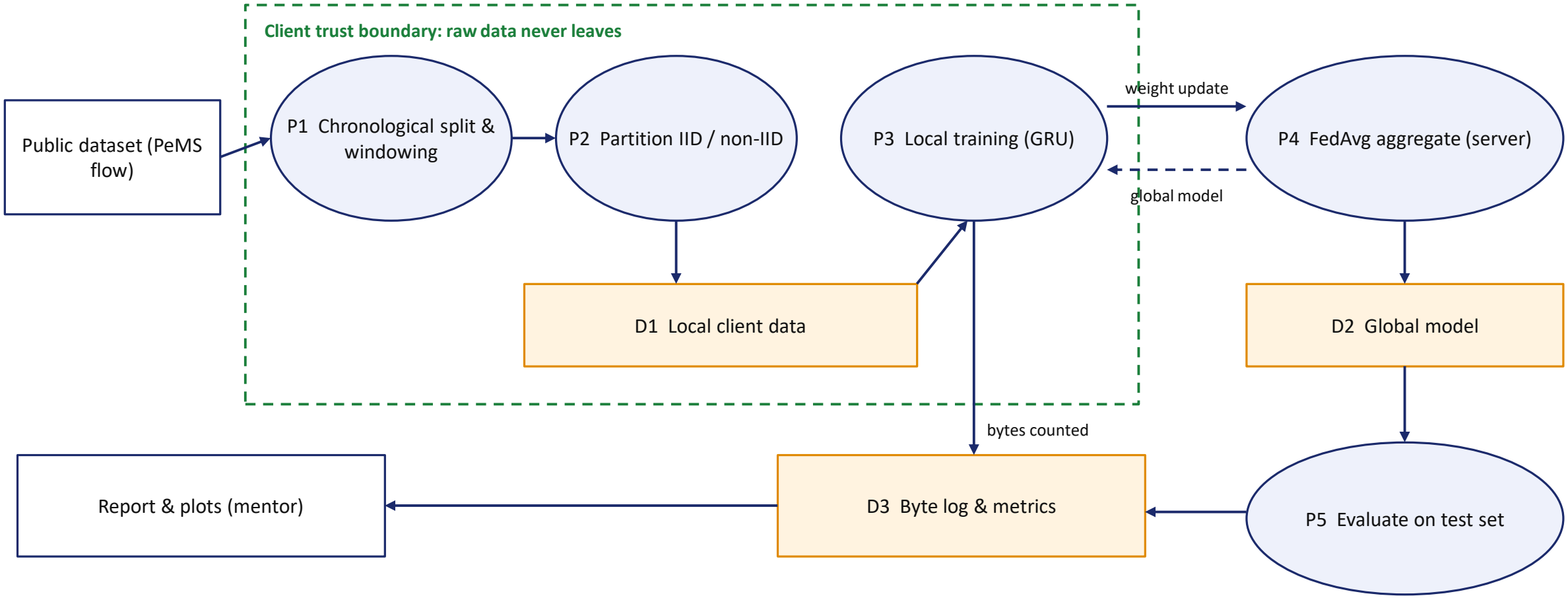


Figure: component view. Solid arrows = upload / output, dashed = download. Raw data stays inside the client.

Data Flow Diagram



Only model weights and byte counts cross the boundary. Raw data stays in D1.

Prototype Pseudocode: FedAvg with Byte Meter

```
for seed in [0, 1, 2]:
    clients = partition(train_data, mode, K)      # IID / non-IID
    w = init_gru(); total_bytes = 0
    for r in 1..R:
        bytes_r = 0                                # per-round
        for k in 1..K:                             # all clients
            w_k = local_train(w, clients[k], E)    # E epochs
            bytes_r += nbytes(w) + nbytes(w_k)     # down + up
        w = sum(n_k * w_k) / sum(n_k)              # FedAvg
        total_bytes += bytes_r
        log(r, rmse(w, val), bytes_r, total_bytes)
    report(test_mae, test_rmse, MB_to_target)
```

Demo plan

Run IID and non-IID side by side. Show the loss curve and a live MB counter per round.

Check before running

Scaler fit on train only. Test period strictly after train period.

Reproducible

Fixed seeds 0, 1, 2. One command regenerates tables and plots.

Deployment Timeline

1 2 3 4 5

Setup & data

- GitHub repo, add aiori-hackathon
- Load PEMS04 flow data
- Chronological split, persistence baseline

Checkpoint: Baseline MAE / RMSE logged

Baselines

- Centralized GRU
- Local-only GRU
- Run with 3 seeds

Checkpoint: Baseline table done

Federated core

- FedAvg simulator (own loop; Flower [8] optional)
- IID and non-IID partitions
- Byte meter per round

Checkpoint: Curves vs rounds

Analysis

- IID vs non-IID comparison
- E = 5 run (optional)
- MB-to-target table and plots

Checkpoint: Tables, mean ± std

Report & demo

- PDF in repo (given structure)
- Pseudocode snippet
- Mentor demo

Checkpoint: Submission ready

Iterative and measurement-driven. Every stage ends with a checkpoint (RMSE, MB per round).

Bibliography

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Thank You for Your Attention!

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